DSA REFERENCE

Breadth-First Search

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Explore a graph in distance layers using a queue, which makes the first visit shortest in an unweighted graph.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

DSA / BFS

DSA

Breadth-First Search

CORE IDEA

Explore a graph in distance layers using a queue, which makes the first visit shortest in an unweighted graph.

Recognition cues

+ minimum number of edges
+ level-order traversal
+ multi-source distance
+ nearest target

Implementation sequence

+ mark on enqueue
+ process by layer when depth matters
+ seed all sources together
+ store parent only if reconstruction is needed

Invariant

The queue contains discovered nodes whose shortest distance is known; later layers cannot improve earlier ones.

Worked reasoning

For rotten oranges or nearest gate, enqueue all sources at distance zero and expand simultaneously.

Complexity

O(V + E) time and O(V) queue/visited space.

Failure modes

- [ ] marking on dequeue and duplicating work
- [ ] using BFS on weighted edges
- [ ] losing layer boundaries
- [ ] starting separate BFS from every source

Practice ladder

Breadth-First Search: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Binary Tree Level Order Traversal**	State the cue, invariant, edge case, and proof before implementation.
2	**Rotting Oranges**	State the cue, invariant, edge case, and proof before implementation.
3	**Word Ladder**	State the cue, invariant, edge case, and proof before implementation.
4	**01 Matrix**	State the cue, invariant, edge case, and proof before implementation.
5	**Open the Lock**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS
REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.